

Inventory-to-sales ratios and the downstream shift of inventory

Part 1 · Why is it important? — solution report with data, method, results and charts

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1. Inventory-to-sales ratios and the downstream shift of inventory

The I/S ratio converts inventory into *months of sales*. A falling ratio means the same sales are supported with less stock (leaner operations); a spike signals that sales dropped faster than production could be cut.

$$\text{I/S ratio} = \text{Inventory (\$)} / \text{Monthly sales (\$)}$$

1.55 → 1.09

Manufacturing I/S 2016 → 2025

65.9 % → 74.7 %

Downstream share of inventory

1.32 → 1.35

Wholesale I/S 2016 → 2025

1.84 (Aug 2020)

Peak I/S (any echelon)

1.62 → 1.76

Retail I/S 2016 → 2025

120

Months analysed

Inventory-to-sales ratio by echelon

Manufacturing grows leaner while downstream echelons carry relatively more stock

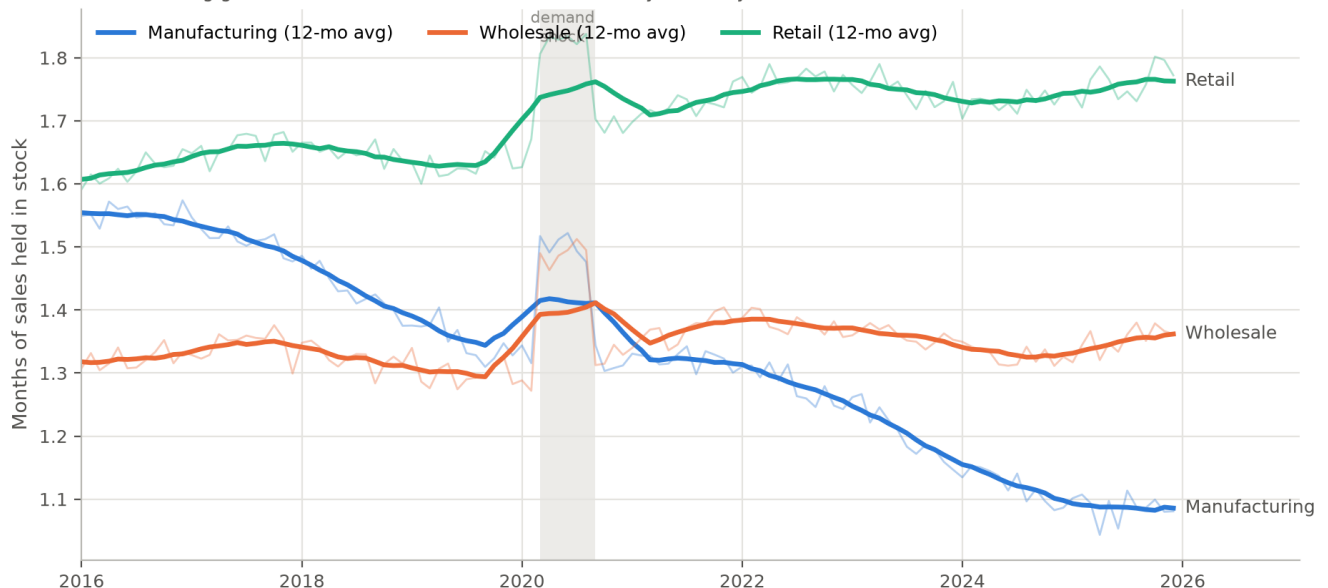


Figure 1.1 — Monthly I/S ratio (thin) and 12-month centred moving average (bold).

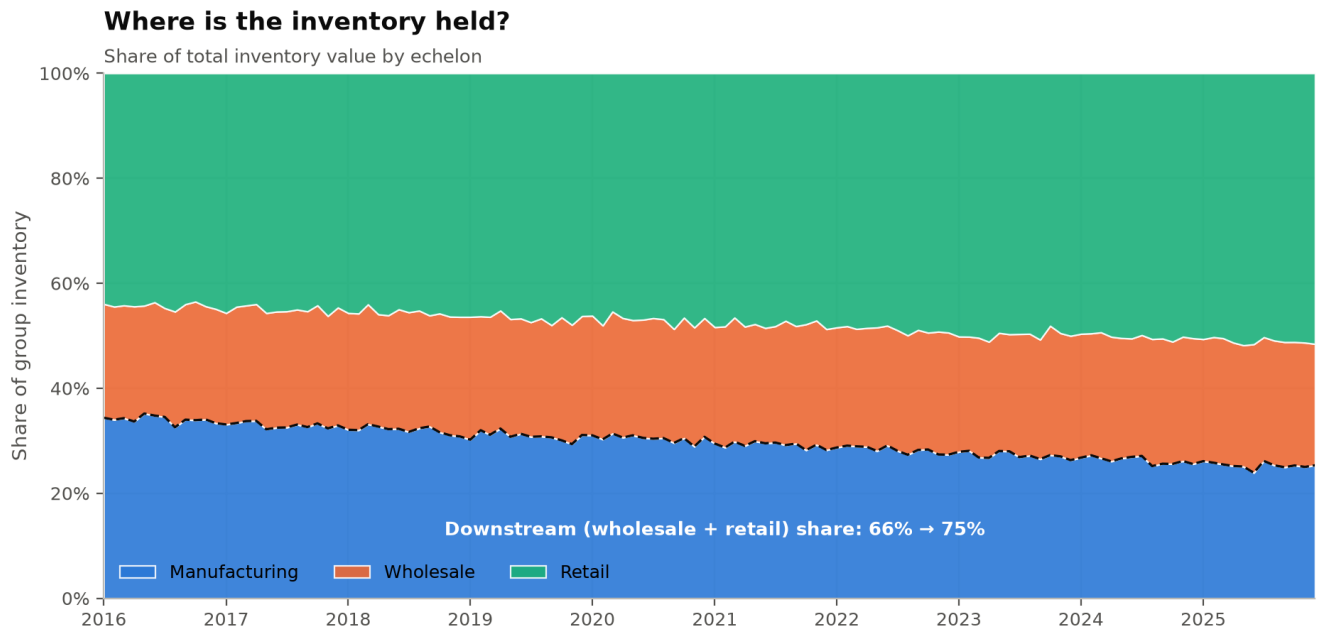


Figure 1.2 — Fraction of total inventory held at each echelon; dashed line = boundary between manufacturing and downstream stock.

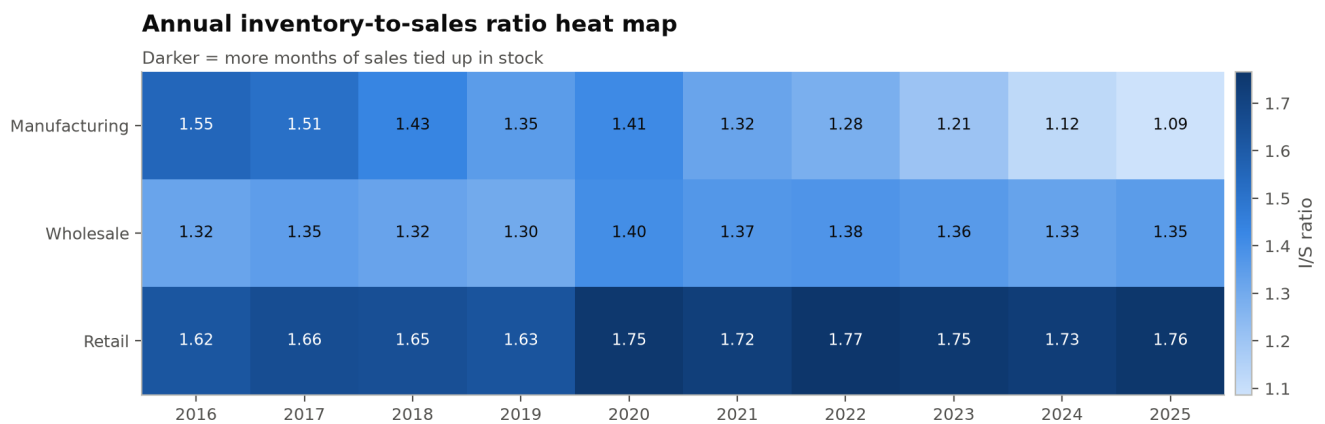


Figure 1.3 — Annual I/S ratio = $\Sigma \text{inventory} / \Sigma \text{sales within the year}$.

Annual I/S ratios and downstream share of inventory.

Year	Manufacturing	Wholesale	Retail	Downstream share %
2016	1.552	1.322	1.621	65.9
2017	1.512	1.345	1.659	67.0
2018	1.432	1.325	1.651	67.9
2019	1.352	1.299	1.629	69.1
2020	1.414	1.402	1.755	69.5
2021	1.323	1.369	1.723	70.8
2022	1.282	1.378	1.766	71.7
2023	1.205	1.360	1.750	72.8
2024	1.122	1.328	1.730	73.7
2025	1.087	1.352	1.760	74.7

- **Global sourcing** lengthens replenishment lead times, so wholesalers and retailers must hold more pipeline and safety stock.
- **Product variety** multiplies SKUs at the point of sale; each SKU needs its own safety stock (see Part 3.2).
- Manufacturers adopting lean / make-to-order practices push finished-goods stock downstream.